

Carissa Mayo

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Professional Summary

Data scientist and quantitative ecologist with experience developing automated, reproducible computational workflows for complex biological and environmental data, with expertise in R, Python, SQL, Bayesian modeling, data integration, and statistical model validation.

Key Skills

Programming Data: R, Python, SQL, Stan, MATLAB, Git, Power BI, Tableau, Excel

Statistical Computational Modeling: Bayesian modeling, machine learning, experimental/observational analysis, model validation, uncertainty quantification, simulation, sensitivity analysis

Education

University of Washington, Seattle Sept 2023 – June 2026
M.S. Quantitative Ecology and Resource Management

University of Colorado Boulder Aug 2019 – May 2023
B.A. Statistics and Data Science, Minors in Computational Biology and Computer Science, Class Rank #1

Experience

Sustainability Data Analyst II May 2026 – present
Deckers Brands – Remote 40 hrs/week

- Manage and analyze environmental and supply-chain datasets supporting value-chain emissions accounting, sustainability reporting, and decision-making.
- Develop and maintain automated workflows in Python and Excel for data processing, validation, and cross-functional reporting.

Graduate Research Assistant June 2024 – May 2026
Pacific Northwest Research Institute – Seattle, WA 40 hrs/week

- Led development of Bayesian population-dynamics models and reproducible R/Stan workflows integrating field observations, laboratory experiments, and environmental data.
- Validated mechanistic models using posterior predictive checks, identifiability assessment, simulation testing, sensitivity analysis, and uncertainty quantification.

Research Assistant, Peleg Lab May 2022 – Aug 2023
University of Colorado – Boulder, CO 20 hrs/week

- Conducted field and laboratory research on honeybee aggregation; collected, curated, and analyzed spatiotemporal biological data with clustering methods.

Laboratory for Interdisciplinary Statistical Analysis Aug 2022 – Dec 2022
University of Colorado – Boulder, CO 10 hrs/week

- Provided statistical consulting support, helping students select appropriate analyses and interpret results.

Data Scientist Intern June 2021 – Aug 2021
Environment for the Americas – Boulder, CO 40 hrs/week

- Conducted data analysis and scientific writing for conservation-focused reporting.

Publications

Giersch, R., et al., including **Mayo, C.** *Variation in Natural Infection Outcomes and Cancer Cell Release from Soft-Shell Clams with Bivalve Transmissible Neoplasia*. **PLOS Pathogens**, 2025.

Gharooni-Fard, G., et al., including **Mayo, C.** *A computational topology-based spatiotemporal analysis technique for honeybee aggregation*. **npj Complex**, 2024.